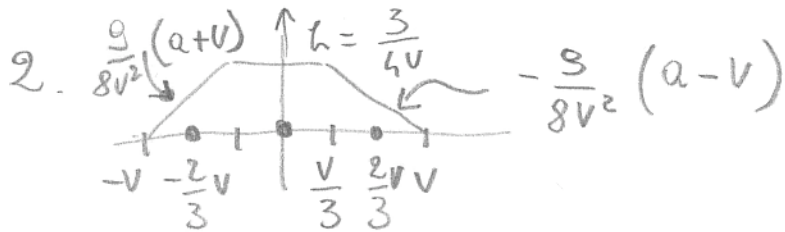


$$1. \quad \sigma_x^2 = \int_0^{\frac{1}{T}} (1 + \cos 2\pi f T) df = \frac{1}{T}$$

$$P[x^2(kT) + x(kT) - 2 > 0] = P[\{x(kT) < -2\} \cup \{x(kT) > 1\}] \\ = P[x(kT) < -2] + P[x(kT) > 1] = \Phi(-2\sqrt{T}) + Q(\sqrt{T})$$



$$\frac{(B_H + b_m)}{2} h = 1 \Rightarrow h = \frac{3}{4V}$$

$$M_e = \int_{-V}^{-\frac{V}{3}} \left(a + \frac{2}{3}V\right)^2 \left(\frac{9}{8V^2}(a+V)\right) da + \\ \int_{-\frac{V}{3}}^{\frac{V}{3}} a^2 \cdot \frac{3}{4V} da + \int_{\frac{V}{3}}^V \left(a - \frac{2}{3}V\right)^2 \left(-\frac{9}{8V^2}(a-V)\right) da \\ = \frac{V^2}{27}$$

3. $c(kT) = a^2(kT)$ e $\bar{e}$ e simboli indipendenti. Inoltre $c(kT) \in \{0, 1\}$, $P[c(kT) = 0] = P[e(kT) = 0] = 0.2$.
 $P[c(kT) = 1] = 0.8$. $\sigma_c^2 = M_c - m_c^2$ $M_c = 0.8$, $m_c = 0.8$

$b(kT)$ viene ottenuto da $c(kT)$ con il filtro

$$h(kT) = \delta_{2CT}(kT) - \delta_{2CT}(kT - T)$$

$$R_b(f) = R_c(f) |H(f)|^2 = \left(\sigma_c^2 \cdot T + m_c^2 \sum_k \delta_R\left(f - \frac{k}{T}\right) \right) \cdot \\ \cdot |1 - e^{-j2\pi f T}|^2 = 0.16T (2 - 2 \cos 2\pi f T)$$

da parte e voglio sommare in punto

$$(2 - 2 \cos 2\pi f T) \cdot \delta_R\left(f - \frac{k}{T}\right) = (2 - 2 \cos 2\pi \frac{k}{T} T) \delta_R\left(f - \frac{k}{T}\right) = 0$$

$$4) \quad c(t) = A_M \cdot f * h(t)$$

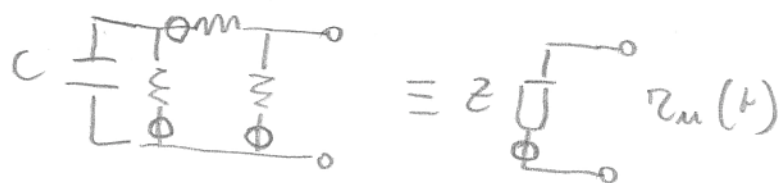
$h(t)$ e $f(t)$ hanno estensione $(-\frac{T}{2}, \frac{T}{2}) \Rightarrow c(t)$ ha estensione $(-T, T)$, non c'è isi.

$$\begin{aligned} c(0) &= A_M \cdot \int_{-\infty}^{+\infty} f(\tau) h(-\tau) d\tau = A_M \cdot h_0 V_0 \cdot \int_{-\infty}^{+\infty} \text{triangle}^2\left(\frac{2\tau}{T}\right) d\tau \\ &= A_M h_0 V_0 \cdot 2 \cdot \frac{T}{2} \cdot \frac{1}{3} = \frac{1}{3} A_M h_0 V_0 T = V'_0 \end{aligned}$$

$$\sigma_m^2 = R_0 \cdot \int_{-\infty}^{+\infty} |H(f)|^2 df = R_0 \int_{-\infty}^{+\infty} h^2(t) dt = \frac{R_0 h_0^2 T}{3}$$

$$P_e = \frac{3}{2} R \left(\frac{V'_0}{\sigma_m} \right)$$

5) Teorema di Nyquist



$$Z = ((R // C) + R) // R$$

$$R // C = \frac{R}{1 + j 2\pi f R C}$$

$$Z = \frac{\left(\frac{R}{1 + j 2\pi f R C} + R \right) R}{2R + \frac{R}{1 + j 2\pi f R C}}$$

$$= \frac{2R + j 2\pi f R^2 C}{3 + j 4\pi f R C}$$

$$R_{en}(f) = 2kT \cdot \text{Re}[Z(f)] = \frac{6R + 8\pi^2 f^2 R^3 C^2}{9 + 16\pi^2 f^2 R^2 C^2} 2kT$$